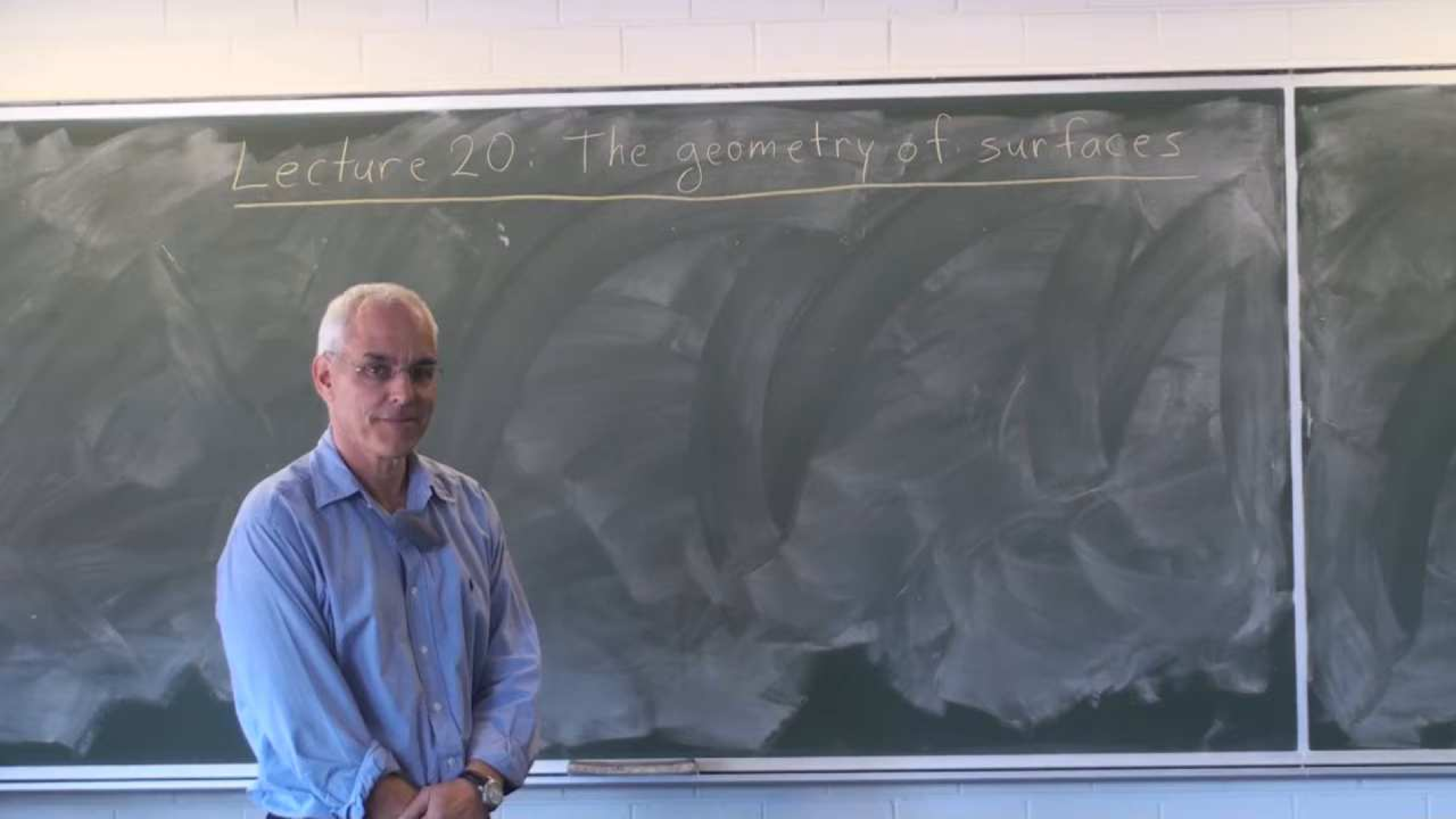
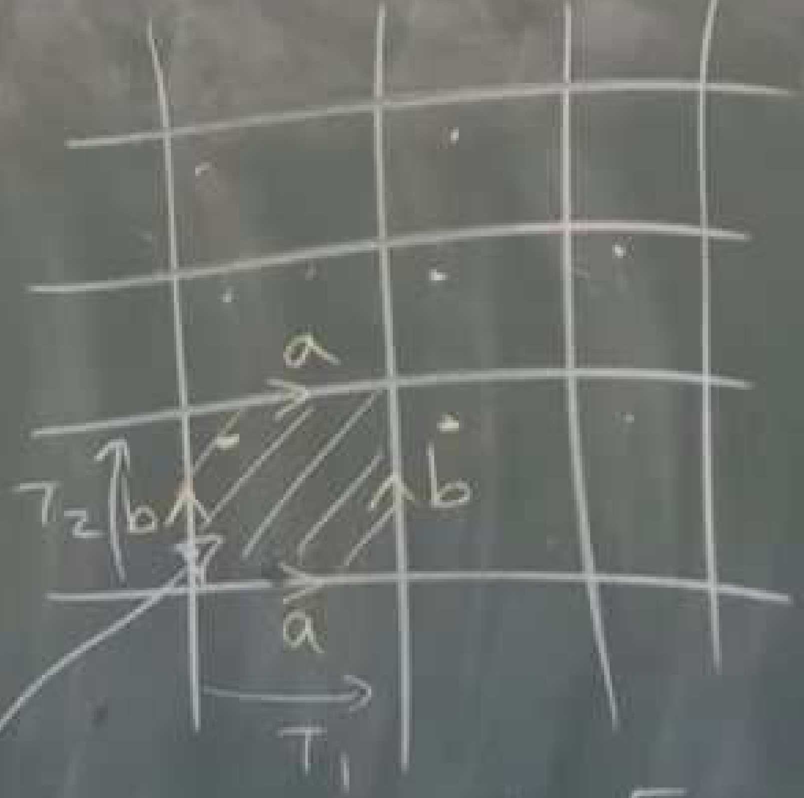
Algebraic Topology: a beginner's course

This lecture:

AlgTop20: The geometry of surfaces


Lecture 20: The geometry of surfaces

Recall.



$T = \mathbb{A}^2 / \langle T_1, T_2 \rangle$

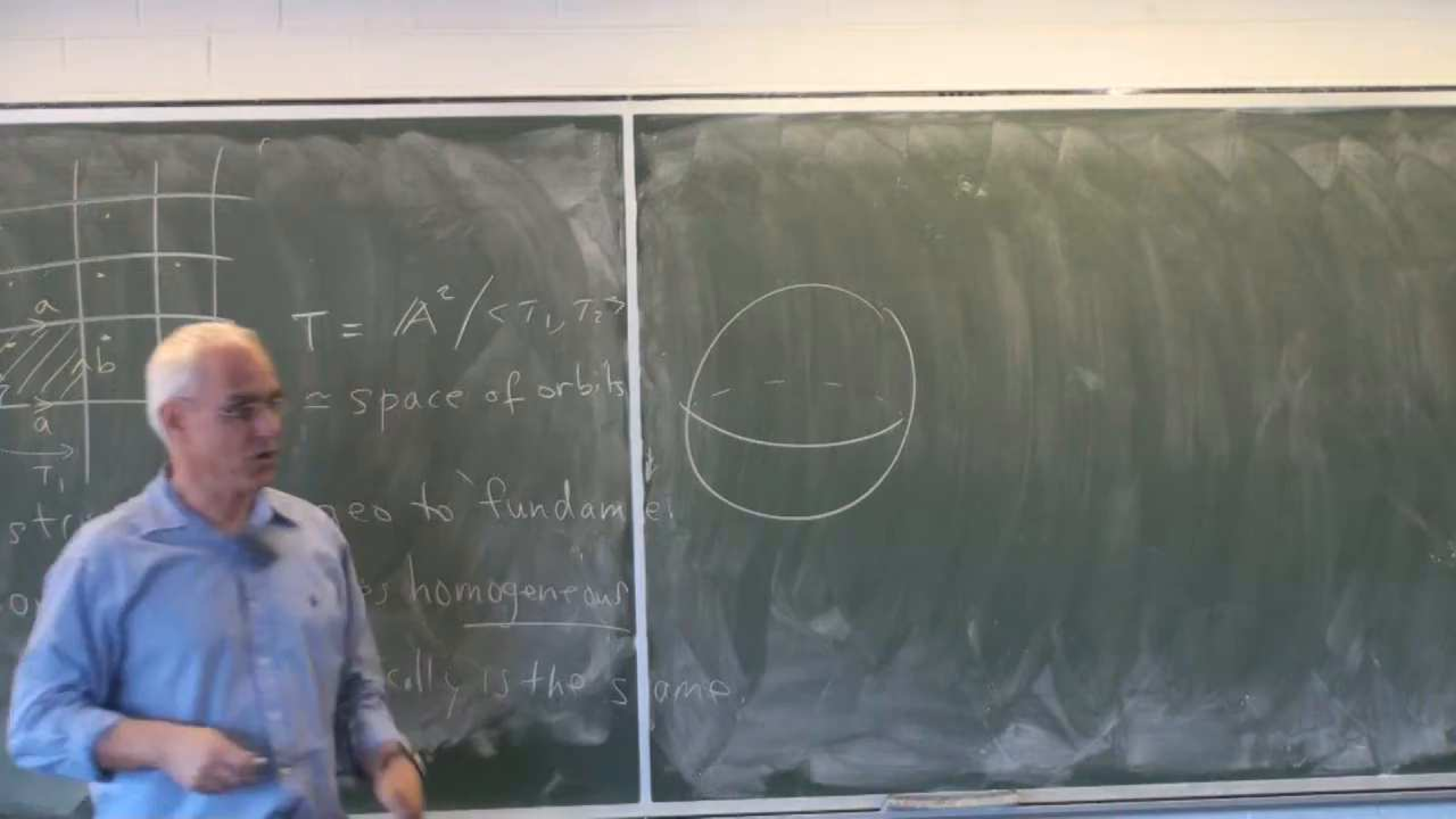
$\simeq$ space of orbits

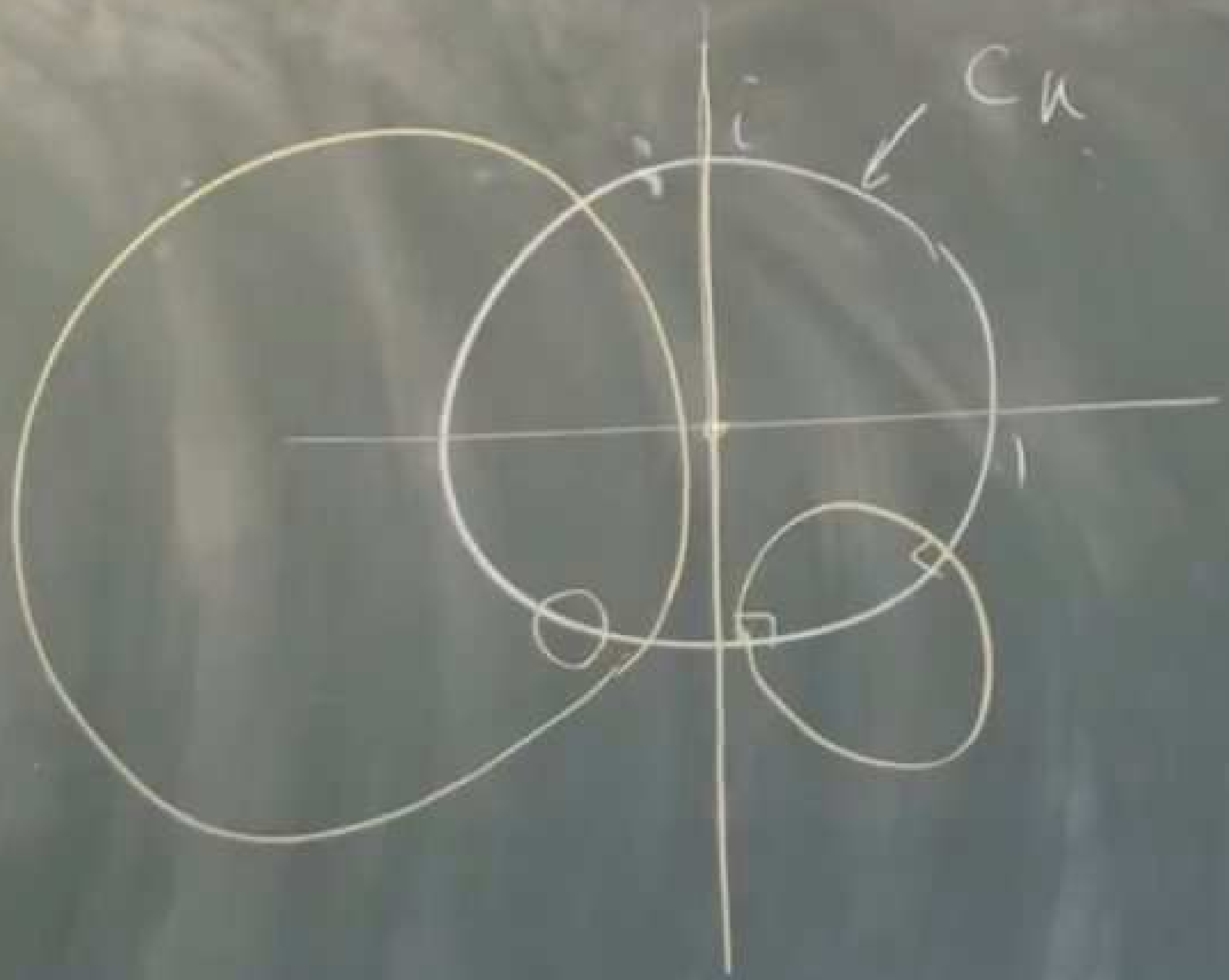
restrict Eucl. geo to 'fundamental domain' : becomes homogeneous

ie every point geometrically is the same.

- symme

Euclide


a
b
a
T_1
$T = \mathbb{A}^2 / \langle T_1, T_2 \rangle$
$\simeq$ space of orbits
fundam
homogeneous
is the same

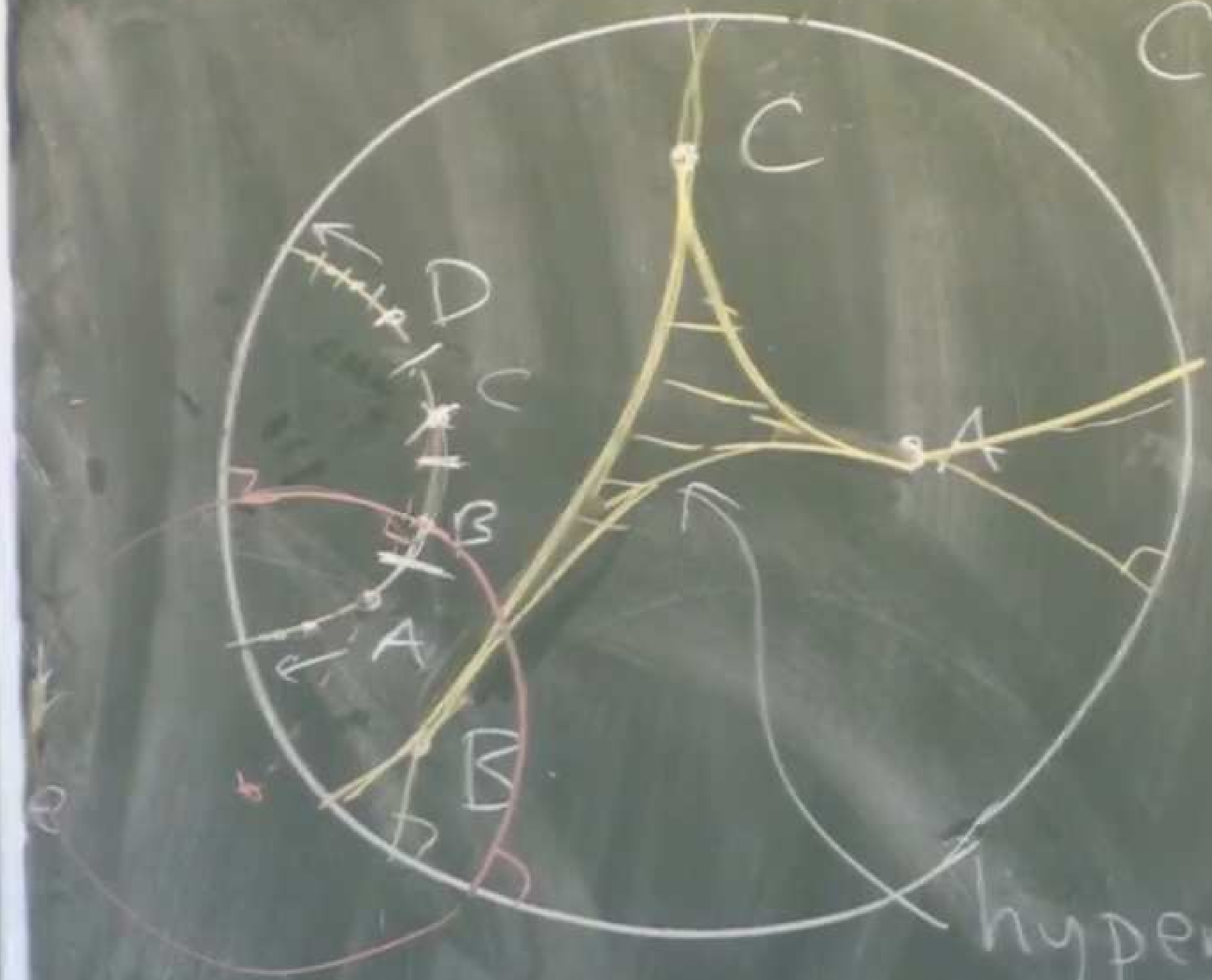


Fix unit circle C_u in $\hat{\mathbb{C}}$.

Consider inversions in circles* perpendicular to C_u.

Analyt

Cu

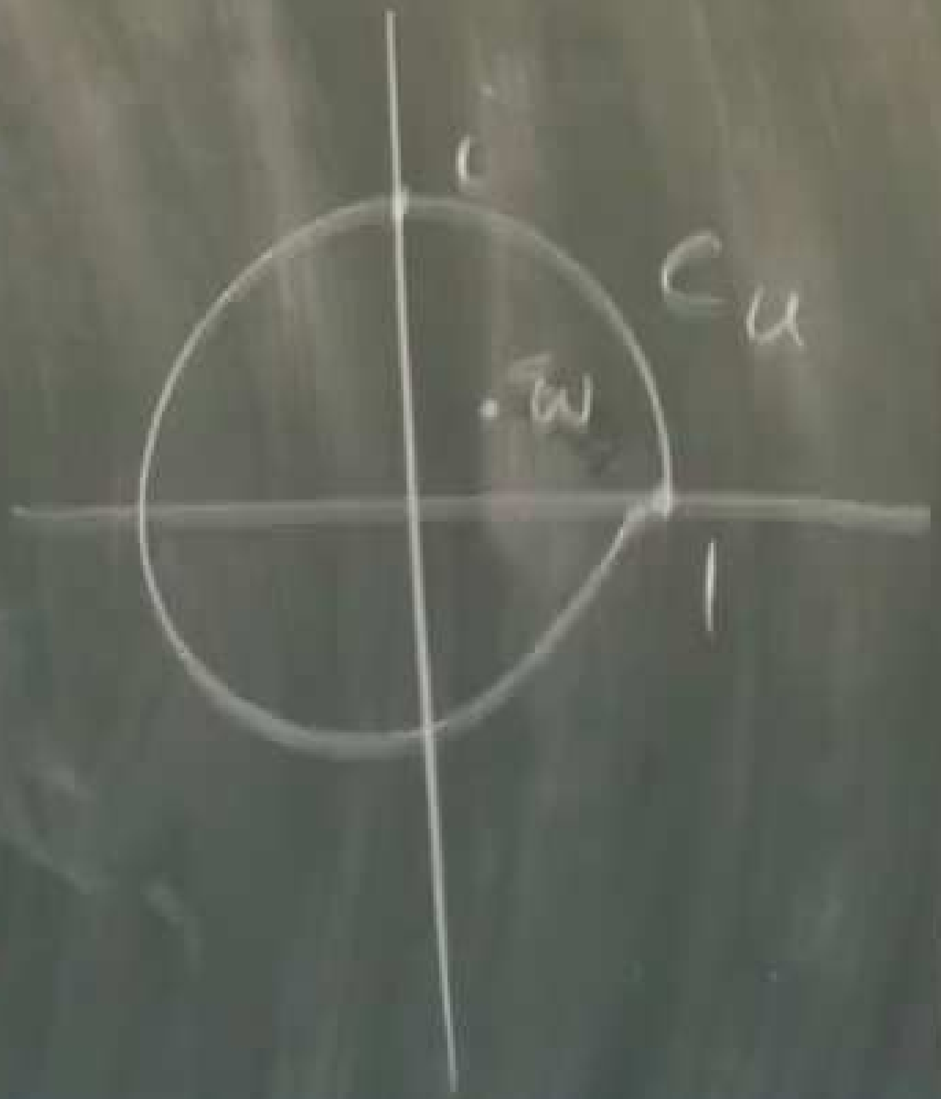


hyperbolic triangle

ABC

Analytic representation of reflections

$\mathbb{C}$



Fix w in $\mathcal{H}$ (inside of di

$$\phi(z) = \frac{z - w}{1 - \bar{w} z}$$

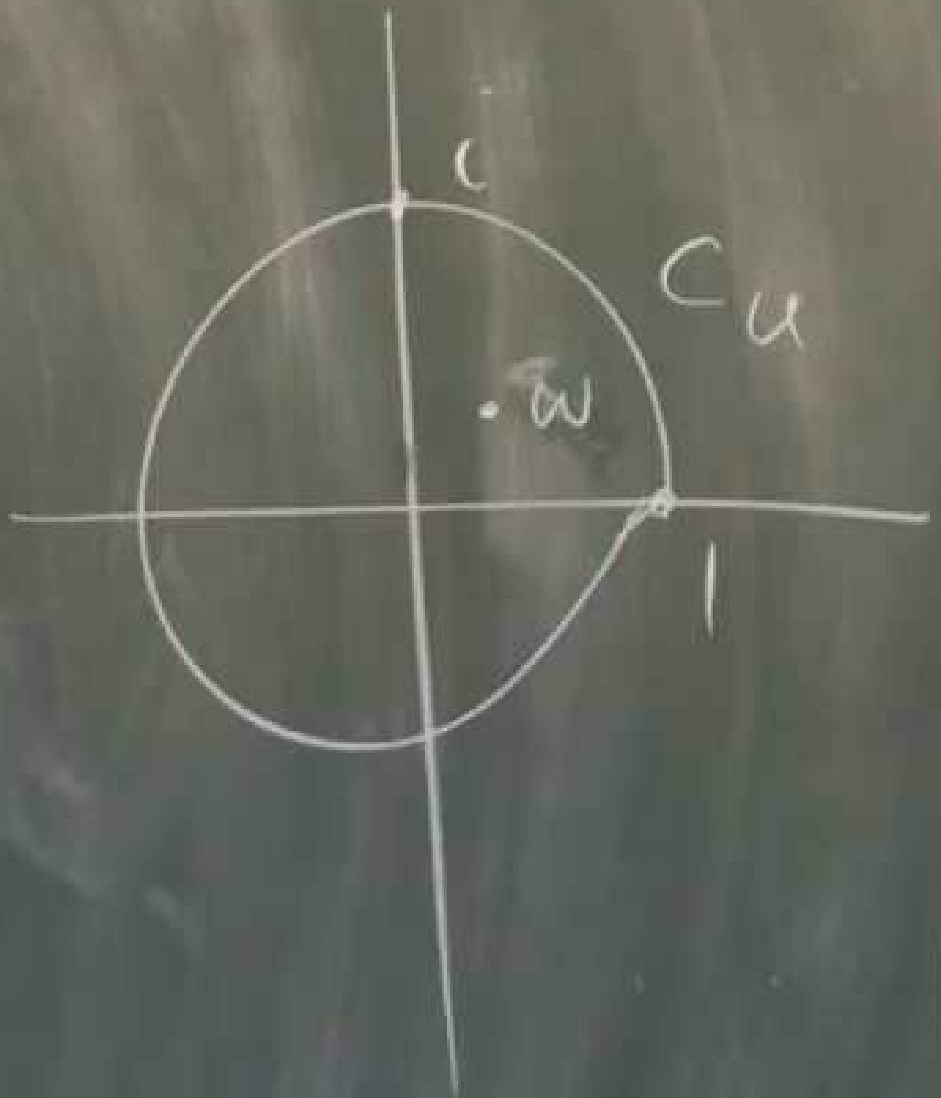
[special case of $f(z) = \frac{az + b}{cz + }$

fractional linear

triangle

Analytic representation of reflections

$\mathbb{C}$



Fix w in $\mathcal{H}$ (inside of ... disk)

$$\phi(z) = \left(\frac{z - w \ldots}{1 - \ldots} \right. \ldots (h)$$

[special case...

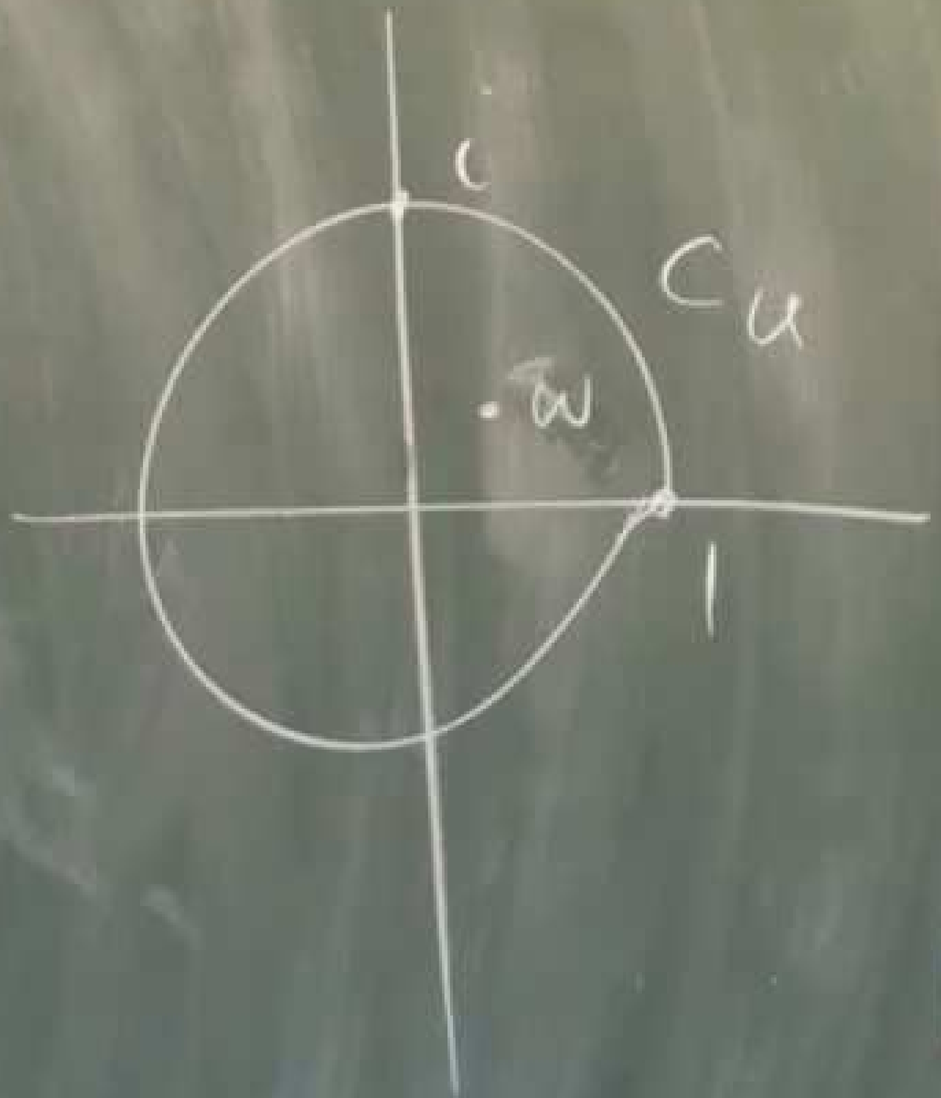
fractio...

M...

triangle

Analytic representation of reflections

$\mathbb{C}$



triangle

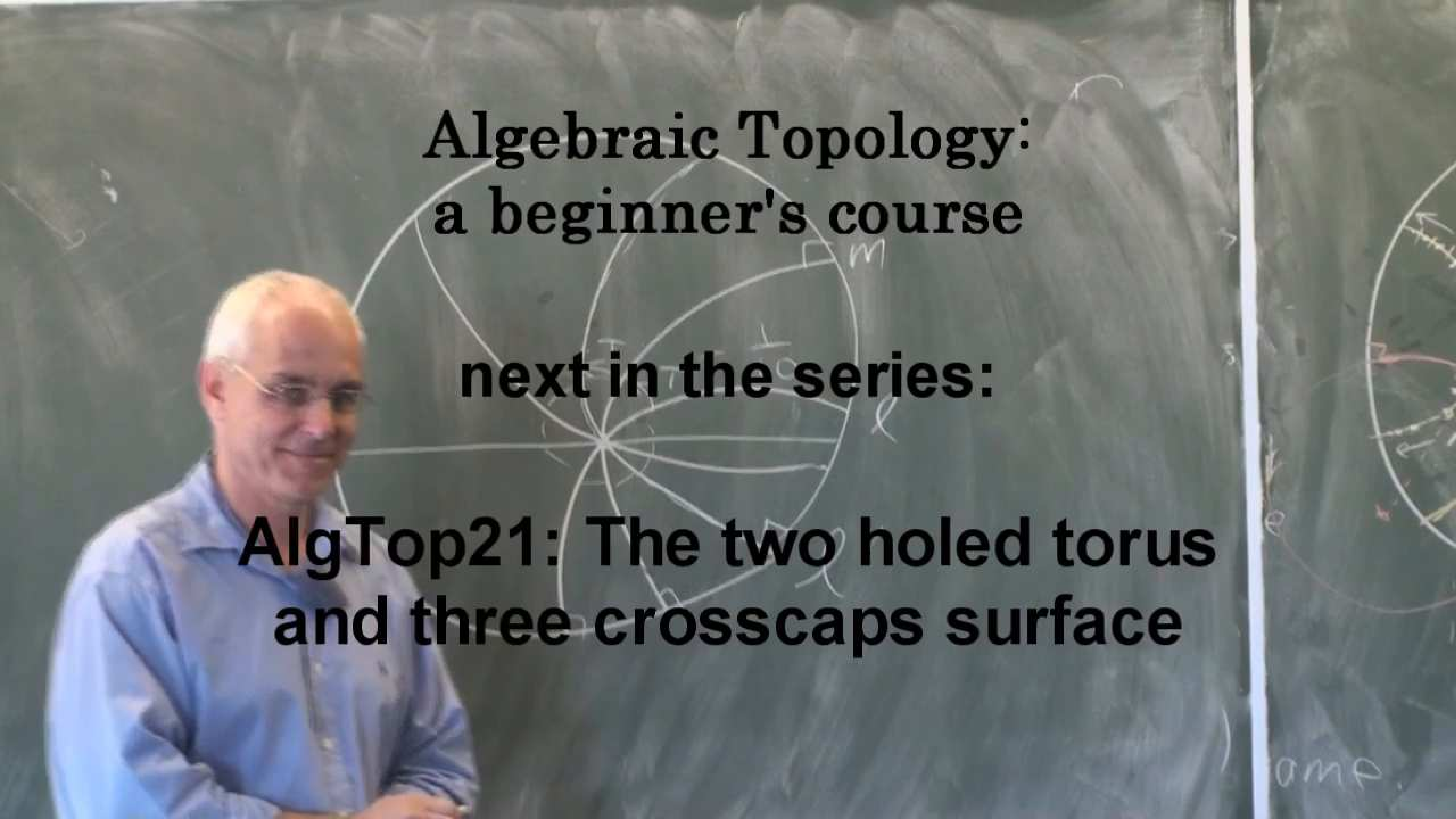
Fix w in $\mathcal{H}$ (inside of disk)

$$\phi(z) = \left(\frac{z - w}{1 - \bar{w} z}\right) e(h)$$

unit circle

[special case of $f(z) = \dfrac{az+b}{cz+d}$]

fractional linear transf. / Mobius transformation


Algebraic Topology:
a beginner's course
next in the series:
AlgTop21: The two holed torus
and three crosscaps surface

Algebraic Topology:
a beginner's course

next in the series:

AlgTop21: The two holed torus
and three crosscaps surface